

A Unified Model for Oral Reading Fluency and Student Prosody

Yihao Wang¹, Zhongdi Wu¹, Joseph Nese², Akihito Kamata¹, Vedant Nilabh¹, Eric C. Larson¹

¹Southern Methodist University, Dallas, TX, USA

²University of Oregon, Eugene, OR, USA

Abstract—In education, analyzing student oral reading is critical to reading comprehension, vocabulary development, and fluency. Automating these assessments is a crucial part of language research—however, researchers often use a separate model for each assessment. Transferring knowledge between assessments can make a model more reliable and robust. In this work, we introduce a unified model comprising a contrastive self-supervised embedding model and a transfer learning suffix network. The embedding model is trained on oral reading fluency data, generating student voice embeddings. Subsequently, the transfer learning network employs these embeddings as features to estimate oral reading prosody scores. Our model demonstrates high agreement with human raters, with a reliability of 0.55 (78% accuracy) on unseen passages compared to 0.63 between humans. This generalization suggests that the model can provide robust representations across various oral reading assessment tasks.

Index Terms—speech recognition, prosody estimation, computational paralinguistics

I. INTRODUCTION

Foundational models in deep learning extract meaningful features that can be transferred to numerous different applications [1]. These models help to accelerate performance in a wide array of tasks, especially classification tasks without large labeled datasets [2]. However, a foundational model does not yet exist for analyzing student speech in an educational context, despite previous works that have shown oral reading to be predictive of many student attributes such as language ability [3], confidence [4], and expressiveness [5]. There are unique challenges to analyzing student oral reading such as mumbling, pitch variation, and odd metre, which are further complicated by the variability among students and the way reading mistakes manifest [6]. Thus, in this work we propose a unified model that is capable of performing multiple oral reading assessment tasks: oral reading fluency and prosody correctness in oral reading. Unifying these tasks into one model is an important first step towards creating a foundational model of student reading.

In this work, we focus on Prosody in the context of oral reading, which is influenced by rhythm, meter, and intonation in spoken words that a student reads from a known passage. Prosody is an important component of spoken communication and is often used as an indicator of reading ability for students [7]. Automated prosody classification has been investigated in applications outside of oral reading including device activation [8] and human-computer interaction [9]. However, we focus on prosody in reading, with related work in language proficiency assessments [10]–[12] and reading proficiency assessments [13]–[15]. Within the realm of assessment of students' language proficiency, Sammit et al. assembled a dataset that employs a four-point scale metric for prosody from NAEP [16], measuring the correctness of prosody as a part of a holistic measure of oral reading fluency [12]. Traditional approaches often involve dataset-specific feature extraction for each dataset and application [17]–[20]. However, without a foundational model such methods may struggle with generalization, hindering their applicability across different contexts. For example, Sammit et al. achieved up to 0.84 Inter-Rater

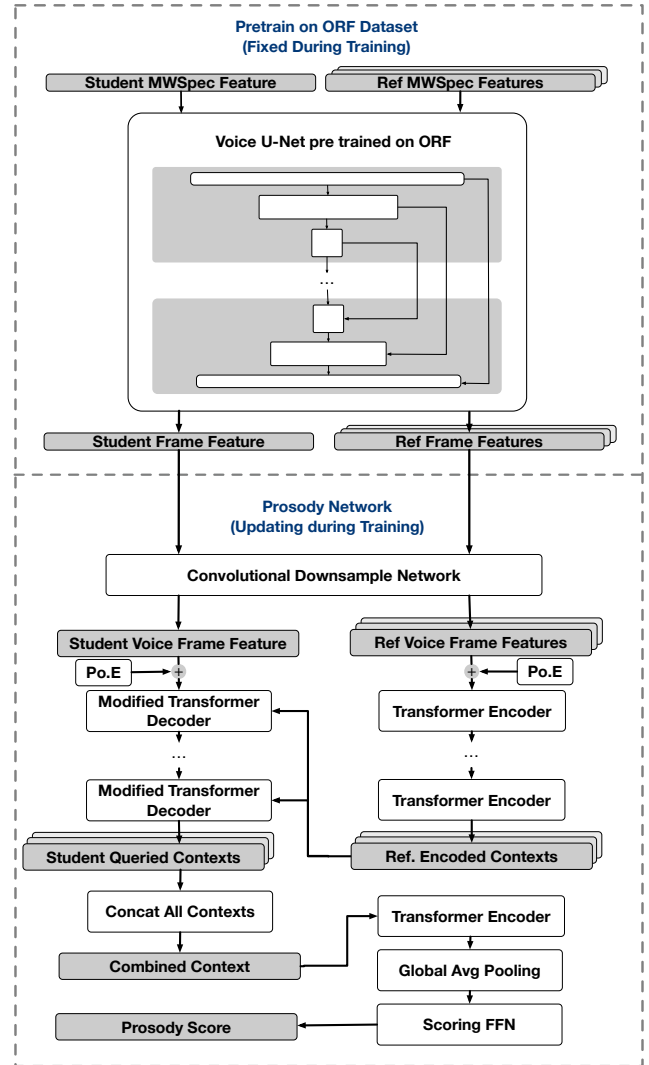


Fig. 1. Overview of our Unified Neural Network Model

Reliability (IRR) of student prosody when testing on known passages, but observed significant degradation when testing on unseen passages, with the highest IRR of 0.46, despite human agreement remaining high [12].

Another common assessment task is to evaluate students' Oral Reading Fluency (ORF) using the metric of words read correctly per minute (WCPM) [21]. Due to the time-consuming assessment of manual WCPM with human listeners, automatic speech recognition (ASR) is often employed for automatic assessment [22]. However, these approaches encounter a variety of problems including low word

recognition performance and nonstandard pronunciation or intonation [23]. Thus, people adopt “passage-specific” methods for ORF assessment, which can learn to handle these challenges in specific passages but struggles to generalize on passages with complex vocabulary [24]. Meanwhile, WCPM and prosody are closely entangled [25], [26]. WCPM encapsulates the fundamental aspects of ORF, focusing on the speed and accuracy. In contrast, prosody extends into more profound contents with not only the accuracy and speed, but also the expressive nuances that reflect deeper comprehension and engagement. Previous works have found that each quantity is related but also distinct [27], [28]. Recognizing the interconnected nature of the two metrics, we explored the development of a unified model taking advantage of their relation, validating generalization ability while accommodating their differences.

In previous work, we proposed a model that adopts text-to-speech (TTS) and contrastive learning to generate acoustic word embeddings of student speech [29]. Notably, the model is trained with unlabeled data with known passages, presenting a unique approach to generate acoustic embeddings and word-level voice embeddings that are passage-agnostic. Building upon the concept of unsupervised acoustic embedding, we incorporate it into this work by integrating transfer learning to estimate a prosody score. Results indicate that our unified model exhibits superior performance in assessing prosody scores on unseen passages, achieving an IRR of 0.55, compared to the human-human IRR of 0.63. Moreover, the model demonstrates a substantial improvement over the previous state-of-the-art model that employed data specific methods, achieving an IRR of 0.46 [12].

II. DATASET

This study involves two datasets. One dataset we refer to as the “prosody dataset” from [12] and is comprised of students reading passages with human evaluated scores of prosody (on a scale of 1-4 from the 2002 NAEP standards [16]). Another dataset we refer to as the “ORF dataset” from [29], comprised of students reading passages and human listeners evaluating WCPM.

A. Prosody Dataset

The student Prosody dataset consists of 6,091 audio recordings in 48 kHz PCM WAVE format, each assessed by at least two human listeners. Students in grades second through fourth grade were recorded reading known passages aloud. Recordings were reviewed by 138 groups of raters paired randomly. Total amount of individual raters is approximately 50. Raters scored prosody on a four-point prosody scale, with the option to score between if they were not certain (1, 1.5, 2, ...) thus creating a seven-point scale start from 1 to 4 with 0.5 increments. Independent scores were averaged, values at or below 0.5 are rounded down, while those above 0.75 are rounded up, ensuring a precise alignment with the four-point integer scale for the final score. Student recordings with a score difference among human raters exceeding 1.0 were considered unreliable. Consequently, two IRR statistics are presented: one, the “Reliable set” excludes all unreliable data, resulting in a human-human IRR of 0.63—the other includes all data, yielding a human-human IRR of 0.58. The Full set consists of 6,091 recordings and the Reliable set consists of 5,838 recordings. For the model training process, approximately 20 percent of recordings are held out as test set. Constraints ensure that each individual student appears exclusively in either training-validation set or test set. During cross-domain analysis, where certain passages are held out, these constraints persist to prevent data leakage.

B. ORF Dataset

The ORF Dataset contains similar passages to those from the Prosody Dataset, but is read by different students in different schools [29]. It initially included 14,339 entries from 662 unique students. However, to maintain quality and relevance, entries shorter than 10 words were eliminated, refining the dataset to 13,029 recordings from 653 distinct students, highlighting the presence of 1,674 unique words within the dataset. Each recording is recorded in 16kHz PCM WAVE format. The ORF Dataset is exclusively audio-based, containing the passage a student was instructed to read alongside a human-evaluated metric of WCPM for evaluation.

III. METHODOLOGY

To address the previously mentioned challenges, we utilize a pre-trained Voice U-Net from [29] to extract frame-based embeddings from both the student’s voice and the voice generated by Text-to-Speech (TTS) technology. By incorporating a reference voice through TTS, we incentivize the model to learn prosody through comparison of the student and reference voice, rather than learning prosody for a specific passage. Thus, we hypothesize that the model will more effectively generalize to passages not included in training. Following the Voice U-Net, a convolutional downsample network is applied to decrease the frame resolution and produce smaller patches. Subsequently, a Transformer-like architecture is employed to align the student’s voice with the reference voice, leading to the generation of the Prosody Score, as shown in Figure 1.

A. Audio Preprocessing

The first step in our methodology is to tailor the audio data for a focus on vocal information, which involves downsampling the original audio to a sampling frequency of 16kHz. We then calculate the 80-bin Mel-Weighted Spectrum (MWSpec) within a frequency range from a lower edge of 80Hz to an upper limit of 7.6kHz. To accomplish this, the Short Time Fourier Transform is employed with a frame length of 1024 samples and a frame step of 128 samples. This procedure generates 80 bins for each frame, where each frame encapsulates a window of 8ms. This spectral analysis is applied to both the student’s and the reference voices, ensuring that the spectral content crucial for vocal analysis is captured.

B. Voice U-Net

The Voice U-Net incorporates a convolutional auto-encoder architecture with skip connections, similar to [30], [31]. These connections are pivotal for reintroducing resolution information at different stages within the network. The U-Net has been trained on ORF tasks, utilizing the same set of passages but with diverse student voices from prosody datasets, which facilitates the generation of frame embeddings [29]. For the prosody assessment task, we input both the MWSpec from the student audio and from four different Siri TTS voices into the network. This process enables us to acquire embeddings for the student’s voice as well as for the TTS references, thus facilitating a comparative analysis.

C. Convolutional Downsampling Network

The Convolutional Downsampling Network aims to reduce the temporal-spatial resolution, reducing computation for later processing by attention-based models. It adopts a Residual Network architecture, further refined with Identity Mapping [32] and replaces Batch Normalization with Layer Normalization to enhance stability [33]. A Dropout layer is added following each weighted layer within the nonlinear segment to prevent overfitting. Average Pooling is

employed to halve the dimensionality after a series of Residual Blocks, effectively reducing the resolution of frame-based features to an eighth of their original size. This reduction results in a more manageable dimensionality of patches, optimizing them for processing by Transformer Attention mechanisms. Subsequently, we introduce Rotary Position Embedding [34], [35] (Figure 1) to the embeddings of both the student's and the reference's outputs, facilitating a nuanced comparison and alignment within the Transformer Attention.

D. Transformer and Attention

Reference Context Encoder: Initially, all reference voices are processed through a series of Transformer Encoders that share weights, utilizing self-attention mechanisms to capture the global context of each frame. This process produces what we refer to as Reference Encoded Contexts, encapsulating the comprehensive contextual information of the reference inputs.

Modified Transformer Decoder: Following this, the downsampled features of the student's voice and each Reference Encoded Context are fed, in pairs, into a shared, modified Transformer Decoder which enables cross-attention between the student and reference features. Deviating from the traditional Transformer framework we omit the Causal Mask on the Attention. This adjustment is pivotal as the objective is not to generate new textual content, but instead to align the reference voice with the student's voice. This alignment is essential, treating the student's voice as the query in comparison to the reference contexts, aiming for a seamless match.

We utilize four Siri TTS voices as references, thus the system produces four distinct pairs of Queried Contexts. Each pair is tailored to capture the nuances of the student's voice in relation to each of the four reference voices. It is important to note that Siri TTS is not intended to serve as a golden standard. Instead, it functions as a consistent reference point, based on the assumption that Siri's reading fluency, prosody, or overall reading style remains stable across different passages. While we used Siri TTS in our setup, it's crucial to emphasize that any modern TTS system that maintains consistency in these aspects could serve as an effective reference. The key requirement is that the TTS used should provide a stable and consistent output during both training and inference (prediction) phases. This allows the model to evaluate the student's voice relative to a controlled and reliable reference, even when the passage is unseen during training.

Output Encoder: In the final step, we concatenate all aligned reference voices and introduce them to an Output Encoder, which employs Attention mechanisms. This strategy is specifically designed to aggregate all aligned references into a unified reference frame, enabling the capture of the global relationships across all frames within the student's voice. This synthesized view facilitates a comprehensive understanding and assessment of the student's voice by comparing it against a singular, consolidated reference standard.

E. Scoring Output Network

Following the Transformer and Attention mechanisms, we implement Global Average Pooling along the time axis, succeeded by a single linear layer Feed-Forward Network (FFN), tasked with generating output scores. These scores are presented in a one-hot format, representing the logits of the probability for each score category.

F. Training Method

The training procedure begins by inputting the MWSpec of both student and reference voices into a pre-trained Voice U-Net, which

then generates embeddings. These embeddings are input into the Prosody Network to generate logits, which are indicative of the prosody scores. The predicted score is compared with 4-point prosody label. Two loss functions are combined into a weighted sum: the cross entropy and mean-squared error of the prediction.

IV. RESULTS

To evaluate the performance of our approach, we designed and tested two configurations of our network: Prosody-L (large) and Prosody-S (small). Both configurations employ the identical downsampling strategy, which leverages embeddings produced by the pre-trained Voice U-Net for each audio frame. These embeddings have a dimensionality of 128, and through the downsampling process, the temporal resolution is reduced by a factor of 1/8, while the embedding dimensionality remains unchanged. The key difference between Prosody-L and Prosody-S resides in the complexity of their Transformer and attention mechanisms. Prosody-L uses four cascaded Reference Context Encoders and four Modified Transformer Decoders. In contrast, Prosody-S adopts a more streamlined approach, incorporating just one Reference Context Encoder and one Modified Transformer Decoder.

Metrics: To comprehensively evaluate our network's performance, we employ three key metrics: accuracy relative to the averaged human score (Accuracy), accuracy within the consensus range of human raters (Consensus Range Accuracy) and Cohen's Kappa (κ) relative to the averaged human score. These metrics are crucial for assessing not only the accuracy of predictions but also the alignment between model's predictions and human assessments, especially in instances of incorrect predictions. Accuracy measures the chance that the model's prediction matches the averaged rater score. Consensus range accuracy calculates the probability that the model's prediction results within the range of two human raters. Finally, Cohen's Kappa, a measure of inter-rater reliability (IRR), introduces a sophisticated understanding of agreement that accounts for chance agreement. According to the literature [36], κ ranges can be interpreted as: 0.4 – 0.5 indicates moderate reliability, 0.6 – 0.8 signifies high reliability, and values above 0.8 denote near perfect reliability. This multifaceted evaluation framework ensures our analysis is aligned with the results section, providing a thorough assessment of network performance through established reliability metrics.

A. Evaluation by Data Split: In- and Cross-Domain

To ensure the generalization capability of both the pre-trained ORF network and the prosody networks, we separate the student voices used during training from those used in the evaluation phase. This approach ensures that neither the prosody model nor the pre-trained Voice U-Net were exposed to the voices in the validation set during training, allowing for a reliable test of the model's ability to generalize to unseen data. Additionally, to ensure that the results are generalizable across the entire dataset, we employed a K-Fold cross-validation method with five folds.

In-Domain: In the "In-Domain" validation approach, we organized the dataset by individual students, using the same passage wording for both the training and validation sets. This method evaluates the model's ability to predict outcomes on data that share the same passages wording but feature different student voices, focusing on the model's proficiency within the same domain as the passages it was trained on. We adopted a 5-fold cross-validation approach, where the train-evaluation sets were split based on students, and the model was trained five times. This method provides a comprehensive evaluation of the network's accuracy and reliability in comparison to human

TABLE I
COMPARISON OF MODELS AND HUMAN PERFORMANCE

Domain / Metrics	Prosody-S	Prosody-L	Human
Mean (Std)			
Cohen κ			
In Domain	0.57 (0.05)	0.57 (0.04)	0.63
Cross Domain	0.54 (0.04)	0.55 (0.06)	
Accuracy			
In Domain	64% (3%)	65% (3%)	63%
Cross Domain	62% (3%)	63% (4%)	
Consensus Range Accuracy			
In Domain	79% (2%)	79% (2%)	/
Cross Domain	77% (2%)	78% (4%)	

Note: Accuracy between humans is the percentage when two raters report the same score.

raters, similar to previous studies. The results, detailed in Table I, show that the Prosody-L model generally outperforms the Prosody-S model on the Reliable dataset. However, in terms of Cohen's κ , neither model achieves human-human agreement levels. As this is a foundational model, our primary concern lies not with In-Domain performance but with the model's generalization capabilities, which are discussed in the next section.

Cross-Domain: To evaluate the model's ability to generalize beyond its training domain, we implemented a "Cross-Domain" validation approach using a Nested K-Fold method. Initially, we divided the students into five folds, as done in the In-Domain experiment. Within each student fold, passages with the same wording were split such that half were reserved for training and the other half for evaluation, ensuring the model was tested on completely unseen wording from passages and from students voices not in training. We employed a stratified method to separate passages in order to maintain a similar IRR levels across both training and validation sets, creating a robust test of the model's generalization.

This Nested K-Fold approach allowed us to rigorously assess the model's generalization capabilities. In total, ten models were trained for this experiment (two for each fold with separate passage wordings). The results, presented in Table I, indicate a slight performance degradation of around 0.02 in κ from "In-Domain" experiment, with Prosody-L outperforming Prosody-S. While both models still fall short of human agreement levels, the results demonstrate a significant improvement of approximately 19.6% over previous studies, where the best reported κ was 0.46 by [12]. Moreover, the accuracy of the models is similar to human agreement.

The difference between the model's accuracy and IRR indicates that although the model achieves similar accuracy to humans, its mistakes are generally more pronounced. When the model errs, its predictions deviate more significantly from the correct scores than human errors, which tend to be closer to the target. This highlights a potential area for enhancing the model's refined understanding and alignment with human judgment.

V. CONCLUSION

This study presented a unified model for automated oral reading prosody and oral reading fluency in an education application, introducing a model that leverages the strengths of a contrastive self-supervised embedding model alongside a transfer learning suffix network. The substantial agreement between our model's evaluations and human raters was evidenced by a Linear Kappa score up to 0.55 on cross-domain passages. Notably, while our model achieves accuracy levels comparable to human evaluators, the degradation in IRR signals a pathway for further refinement.

Moreover, the observed generalization across unseen passages affirms the robustness of the contrastive self-supervised embedding model, and performance remains stable even when passage with unseen wording are used for testing. We did not conduct an extensive hyperparameter search, which could improve generalization.

While our study utilized Siri TTS as a consistent reference point, it is worth noting that the potential impact of different TTS systems or settings was not explored. Future experiments could validate whether our approach holds across various TTS configurations, further strengthening the generalizability of our approach.

In conclusion, our study represents a significant step forward in the automated assessment of oral reading prosody, unifying with a model of ORF. It highlights the generalizability of proposed methodologies while revealing the challenges that remain, particularly the notable error magnitudes when the model makes mistakes, providing a basis for future work to improve the accuracy, reliability, and general applicability of these educational tools. This technology also has the potential to be applied across a broader spectrum of applications, particularly in scenarios where embeddings can be trained on one speech-corpus and be effectively transferred to and utilized for similar tasks within student oral reading.

REFERENCES

- [1] C. Zhou, Q. Li, C. Li, J. Yu, Y. Liu, G. Wang, K. Zhang, C. Ji, Q. Yan, L. He, *et al.*, "A comprehensive survey on pretrained foundation models: A history from bert to chatgpt," *arXiv preprint arXiv:2302.09419*, 2023.
- [2] S. Niu, Y. Liu, J. Wang, and H. Song, "A decade survey of transfer learning (2010–2020)," *IEEE Transactions on Artificial Intelligence*, vol. 1, no. 2, pp. 151–166, 2020.
- [3] A. Johnson, H. Veeramani, B. Natarajan, and A. Alwan, "An equitable framework for automatically assessing children's oral narrative language abilities," in *Proc. Interspeech*, pp. 2023–1257, 2023.
- [4] K. Sabu and P. Rao, "Automatic prediction of confidence level from children's oral reading recordings," in *Proc. Interspeech*, pp. 3141–3145, 2020.
- [5] K. Sabu, K. Kumar, and P. Rao, "Automatic detection of expressiveness in oral reading," in *Proc. Interspeech*, pp. 1489–1490, 2018.
- [6] N. Edo-Marzá, "Pronunciation and comprehension of oral english in the english as a foreign language class: Key aspects, students' perceptions and proposals," *Journal of Language Teaching and Research*, 2014.
- [7] A. Rosenberg, *Automatic Detection and Classification of Prosodic Events*. PhD thesis, Columbia University, 01 2009.
- [8] X. Zhang, Z. Su, and J. Rekimoto, "Aware: Intuitive device activation using prosody for natural voice interactions," in *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, CHI '22, (NY, USA), ACM, 2022.
- [9] S. Z. Razavi, L. K. Schubert, K. van Orden, M. R. Ali, B. Kane, and E. Hoque, "Discourse behavior of older adults interacting with a dialogue agent competent in multiple topics," *ACM Trans. Interact. Intell. Syst.*, vol. 12, jul 2022.
- [10] Y. Zhang, F. Weninger, A. Batliner, F. Hönl, and B. Schuller, "Language proficiency assessment of english 12 speakers based on joint analysis of prosody and native language," in *Proceedings of the 18th ACM International Conference on Multimodal Interaction*, ICMI '16, (NY, USA), p. 274–278, ACM, 2016.
- [11] M. Nicolao, M. Sanders, and T. Hain, "Improved acoustic modelling for automatic literacy assessment of children," in *Interspeech 2018, 19th Annual Conference of the International Speech Communication Association, Hyderabad, India, 2-6 September 2018* (B. Yegnanarayana, ed.), pp. 1666–1670, ISCA, 2018.
- [12] G. Sammit, Z. Wu, Y. Wang, Z. Wu, A. Kamata, J. Nese, and E. Larson, "Automated prosody classification for oral reading fluency with quadratic kappa loss and attentive x-vectors," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 3613–3617, 05 2022.
- [13] M. Duong, J. Mostow, and S. Sitaram, "Two methods for assessing oral reading prosody," *ACM Transactions on Speech and Language Processing (TSLP)*, vol. 7, no. 4, pp. 1–22, 2011.

- [14] D. D. Paige, T. Rasinski, T. Magpuri-Lavell, and G. S. Smith, "Interpreting the relationships among prosody, automaticity, accuracy, and silent reading comprehension in secondary students," *Journal of Literacy Research*, vol. 46, no. 2, pp. 123–156, 2014.
- [15] K. Sabu and P. Rao, "Automatic assessment of children's oral reading using speech recognition and prosody modeling," *CSI Transactions on ICT*, vol. 6, pp. 221–225, 2018.
- [16] S. White, J. Sabatini, B. J. Park, J. Chen, J. Bernstein, and M. Li, "The 2018 naep oral reading fluency study. nces 2021-025.," *National Center for Education Statistics*, 2021.
- [17] H. Kallio, A. Suni, P. Virkkunen, and J. Šimko, "Prominence-based Evaluation of L2 Prosody," in *Proc. Interspeech 2018*, pp. 1838–1842, 2018.
- [18] L. Fontan, M. Le Coz, and S. Detey, "Automatically Measuring L2 Speech Fluency without the Need of ASR: A Proof-of-concept Study with Japanese Learners of French," in *Proc. Interspeech 2018*, pp. 2544–2548, 2018.
- [19] Q.-T. Truong, T. Kato, and S. Yamamoto, "Automatic Assessment of L2 English Word Prosody Using Weighted Distances of F0 and Intensity Contours," in *Proc. Interspeech 2018*, pp. 2186–2190, 2018.
- [20] N. Hu, B. Janssen, J. Hanssen, C. Gussenhoven, and A. Chen, "Automatic Analysis of Speech Prosody in Dutch," in *Proc. Interspeech 2020*, pp. 155–159, 2020.
- [21] T. G. Morrison and B. Wilcox, "Assessing expressive oral reading fluency," *Education Sciences*, vol. 10, no. 3, 2020.
- [22] P. Rao, P. Swarup, A. Pasad, H. Tulsiani, and G. G. Das, "Automatic assessment of reading with speech recognition technology," *Asia-Pac. Society for Comp. in Ed.*, p. 1, 2016.
- [23] J. Tobin and K. Tomanek, "Personalized automatic speech recognition trained on small disordered speech datasets," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 6637–6641, 2022.
- [24] B. B. Klebanov and A. Loukina, "Exploiting structured error to improve automated scoring of oral reading fluency," in *International Conference on Artificial Intelligence in Education*, pp. 76–81, Springer, 2021.
- [25] N. Calet, N. Gutiérrez-Palma, and S. Defior, "Effects of fluency training on reading competence in primary school children: The role of prosody," *Learning and Instruction*, vol. 52, pp. 59–68, 2017.
- [26] R. Hudson, H. Lane, and P. Pullen, "Reading fluency assessment and instruction: What, why, and how?," *Reading Teacher - READ TEACH*, vol. 58, pp. 702–714, 05 2005.
- [27] P. J. Schwanenflugel and R. G. Benjamin, "Lexical prosody as an aspect of oral reading fluency," *Reading and Writing*, vol. 30, pp. 143–162, 2017.
- [28] W.-L. Chung, "Oral reading prosody and the relation with reading abilities: A comparison of two rating scales," *Reading and Writing*, vol. 37, no. 4, pp. 869–886, 2024.
- [29] Y. Wang, Z. Wu, V. Nilabh, A. Kamata, J. Nese, and E. Larson, "Improving oral reading fluency assessment through sub-sequence matching of acoustic word embeddings," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, p. 6, 09 2024.
- [30] R. Giri, U. Isik, and A. Krishnaswamy, "Attention wave-u-net for speech enhancement," in *2019 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, pp. 249–253, IEEE, 2019.
- [31] B. Tolooshams, R. Giri, A. H. Song, U. Isik, and A. Krishnaswamy, "Channel-attention dense u-net for multichannel speech enhancement," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 836–840, IEEE, 2020.
- [32] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *Computer Vision – ECCV 2016* (B. Leibe, J. Matas, N. Sebe, and M. Welling, eds.), (Cham), pp. 630–645, Springer International Publishing, 2016.
- [33] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.
- [34] L. Samarakoon and T.-Y. Leung, "Conformer-based speech recognition with linear nystrom attention and rotary position embedding," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 8012–8016, IEEE, 2022.
- [35] X. Li, S. Li, X.-L. Zhang, and S. Rahardja, "Transformer-based end-to-end speech translation with rotary position embedding," *IEEE Signal Processing Letters*, 2024.
- [36] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.